

Correlation

◦ Relationship between
two variables

→ how strong?

→ the direction?

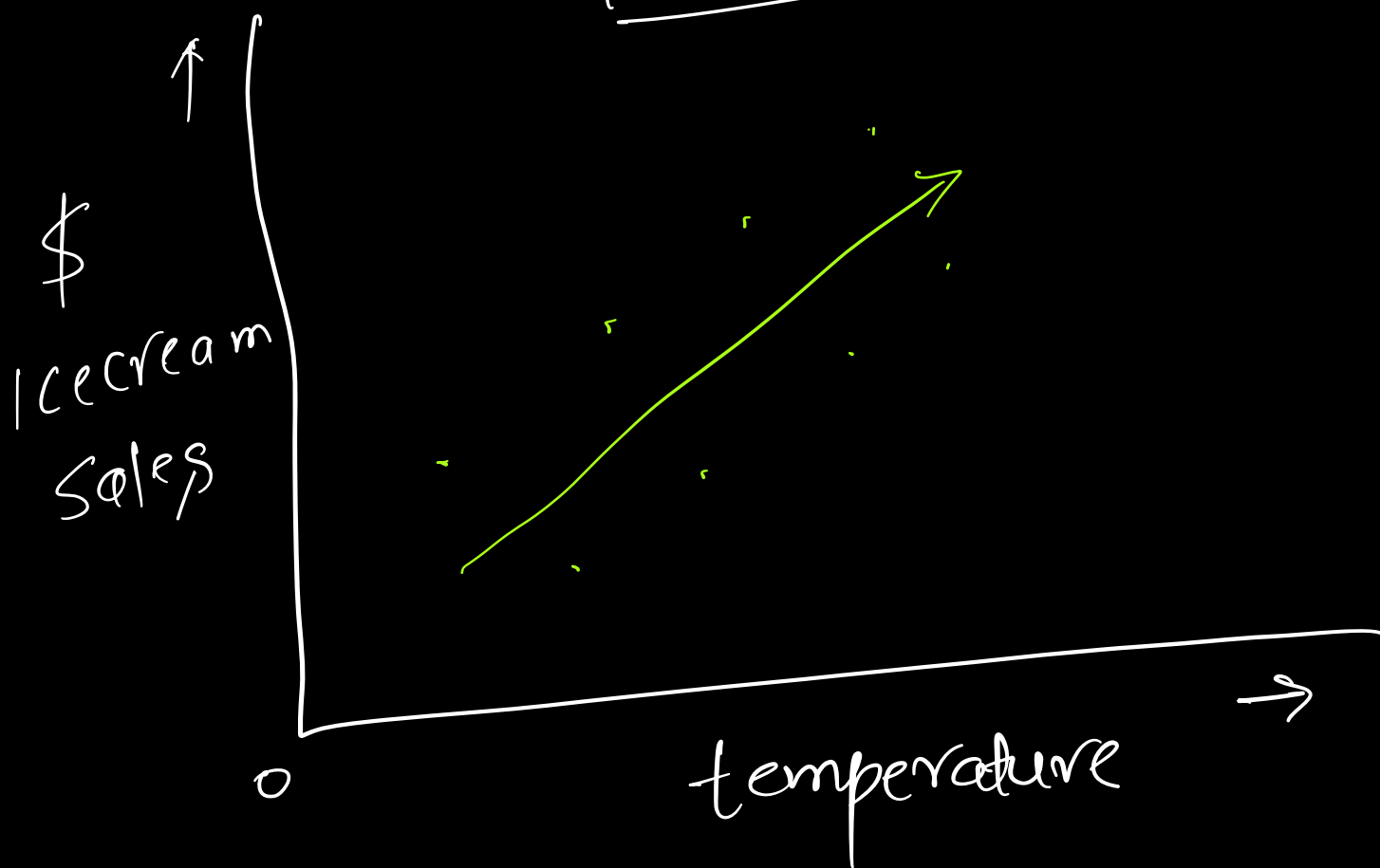
Directions?

1) positive trends

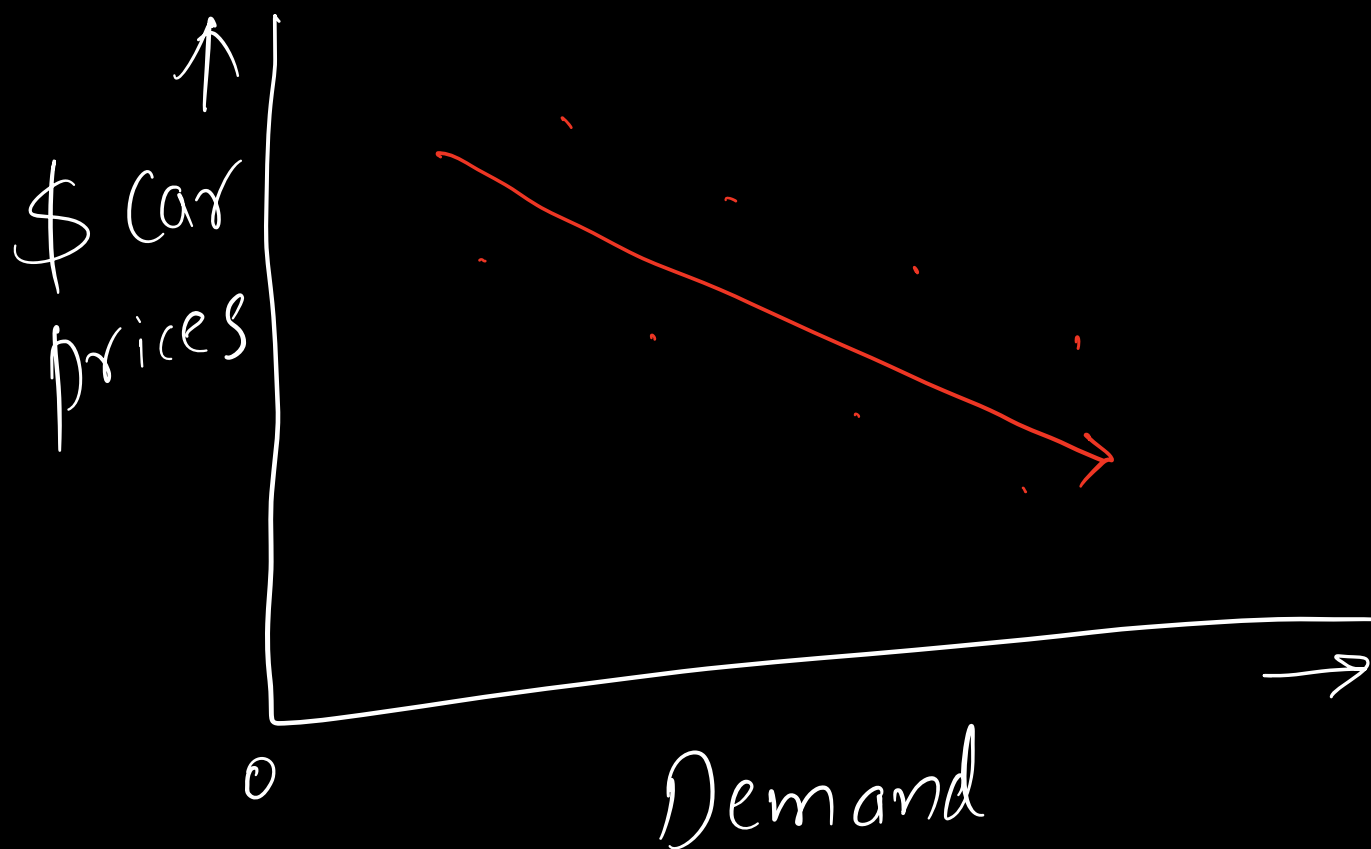
2) Negative trends

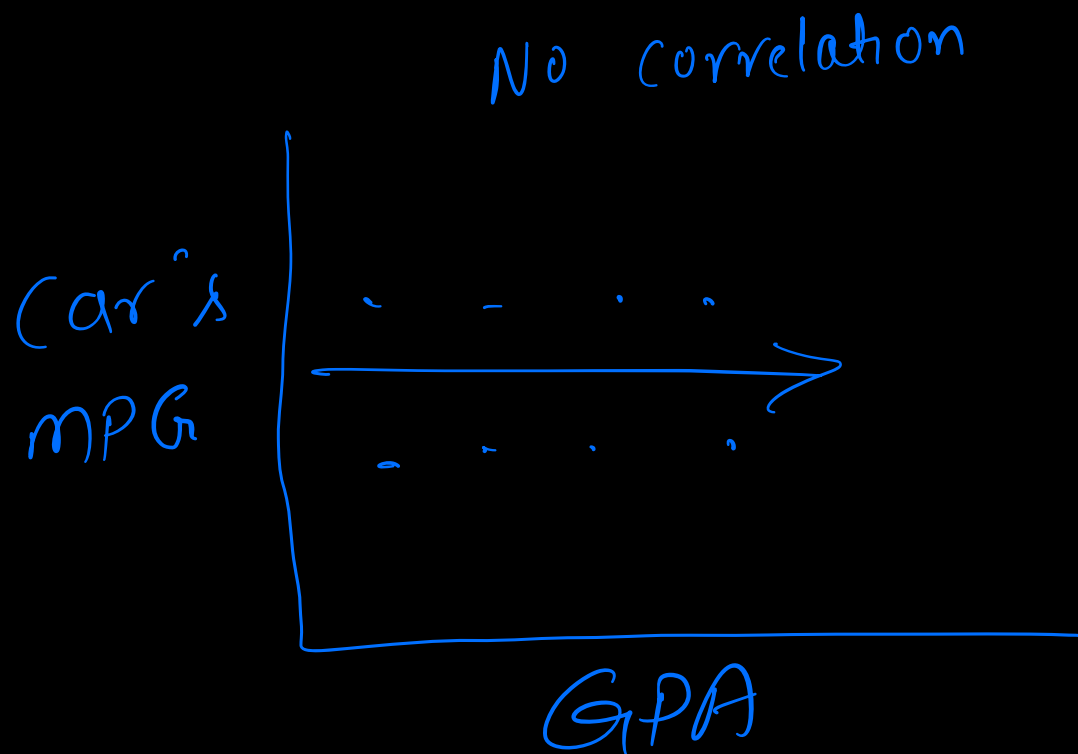
3) No trends (no relationship)

↑ve trend



-ve trend





Variance and standard deviation describe the spread of a single variable, while covariance and correlation measure how two different variables move together

Covariance

To check whether two variables are related or not $\Rightarrow$ do ^{they} covary

→ talks about direction of the linear relationship betⁿ two variables

+ve → moving in the same direction

-ve → " " " " opposite "

Standard Deviation

$$\sigma_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

Variance

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Covariance =
$$\frac{\sum_{i=1}^n (x_i - \bar{x}) \cdot (y_i - \bar{y})}{n}$$

Students hours practice/studied & score

Student	Hrs (x)	Score (y)	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x}) \cdot (y_i - \bar{y})$
A	1	55	-2.5	-12.5	31.25
B	2	50	-1.5	-17.5	26.25
C	3	70	-0.5	2.5	-1.25
D	4	65	0.5	-2.5	-1.25

E	5	80	1.5	12.5	18.75
F	6	85	2.5	17.5	43.75

$$\bar{x} = 3.5 \quad \sum (x_i - \bar{x})^2 = 17.5 \quad \sum (y_i - \bar{y})^2 = 937.5 \quad \sum (x_i - \bar{x}) \cdot (y_i - \bar{y}) = 117.5$$

$$\bar{y} = 67.5$$

$$\text{Covariance}(x, y) = \frac{117.5}{6} = 19.583$$

$-\infty$ to ∞

$$\text{correlation} = \frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y}$$

$$\sigma_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}}$$

$$\sigma_x = \sqrt{\frac{17.5}{6}}$$

$$\begin{aligned}
 &= \sqrt{2.91} \\
 &= 1.707 \\
 s_y &= \sqrt{\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{n}} = \sqrt{\frac{937.5}{6}} \\
 &= \sqrt{156.25} \\
 &= 12.5
 \end{aligned}$$

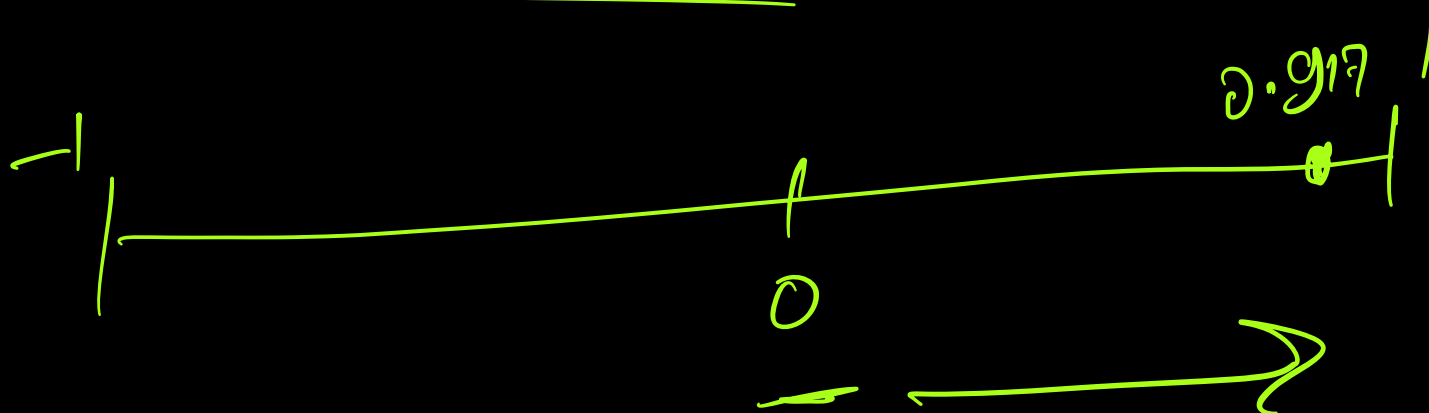
$$\text{Correlation} = \frac{\text{Cov}(x, y)}{s_x \cdot s_y}$$

$$= \frac{19.583}{1.707 \times 12.5}$$

$$1.707 \times 12.5$$

$$\text{Correlation} = 0.917$$

Pearson's
correlation
(r)



Anchor values for correlation

+1 → perfect +ve

-1 → perfect -ve

0 → No correlation

(No linear relationship)

A rough scale

0.0 to 0.3 → weak correlation

0.3 to 0.7 → moderate "

0.7 to 1.0 → strong "

mirror for negative values

(Note: Not law, but convention)

P-value

The p-value is the probability of seeing a correlation at least this strong purely by chance, if there were no relationship at all.

$\Rightarrow$ Small p-value $\rightarrow$ "Chance alone would rarely produce this value (in this case 0.017)"

$\Rightarrow$ large p-value $\rightarrow$ "Chance could easily produce this value, so do not trust this"

P-value ($P < 0.05$) $\Rightarrow$ considered

statistically significant

If $P = 0.01 \Rightarrow$ meaning

If hours and score were truly unrelated, there is only a 1% chance we could see a correlation this strong from 6 random student

This ^{is} low, so the relationship looks real, not a fluke.

Common cut-off 0.05

1. Null hypothesis: No relation

2. Reject null hypothesis:

p-value & Sample Size

$r = 0.5$, $n = 5$ $p = 0.33$ (Not significant)
 $r = 0.5$, $n = 10$ $p = 0.14$ (still not significant)
 $r = 0.5$, $n = 20$ $p = 0.05$ (now significant)
 $r = 0.5$, $n = 100$ $p = 0.001$ (very "
 $r = 0.5$, $n = 1000$ $p = 0.00001$ (extremely "

$\therefore p < 0.05$ is statistically significant
for a correlation